QSE - Quantitative Strategy Engine

A C++ event-driven backtesting & execution-research engine:
how much of a backtest's profit is real, and how much is an artifact of ignoring market microstructure?

github.com/MarcusJMyrick/Quantitative-Strategy-Engine-QSE-

ARCHITECTURE

Data layer
CSV / Parquet readers
Python pipeline
ZeroMQ tick feed

->

C++ engine
SPSC ring -> BarBuilder
Strategies -> OrderManager
Full-Depth Order Book

->

Analysis
equity + tradelog
tearsheet.py
research artifacts

Live mode: same strategy code -> Alpaca paper venue (order + fill reconciliation)

Research track (QR): eigen stat arb | HMM regime -> lambda | meta-labeling | CPCV + Deflated Sharpe + MDA (the judge)

KEY RESULTS

(engine + low-latency, then research track - honest negatives included)

Phantom profit @ 25k sh/signal
\$813,700 (naive Sharpe +1.93 -> real -5.26)

Market-impact exponent
b = 0.569 vs sqrt-law 0.5 (R^2 = 0.999)

Arena allocator vs new/delete
3.5 ns vs 57-70 ns (16-20x)

Lock-free SPSC ring, tail latency
p99 42 ns vs 16,334 ns (389x); TSan-clean

Cross-platform determinism
Docker == native: Sharpe -2.404, 456 trades

Live paper trading (Alpaca)
5 signals -> 5 fills -> 5/5 reconciled

Stat arb vs momentum, Engine B
credible negative: momentum 0.84 > stat arb 0.69

Deflated Sharpe (multiple testing)
100 noise: PSR 0.99 but DSR 0.47

HMM regime overlay
calm 44% / elevated 33% / turbulent 23% -> A5 lambda

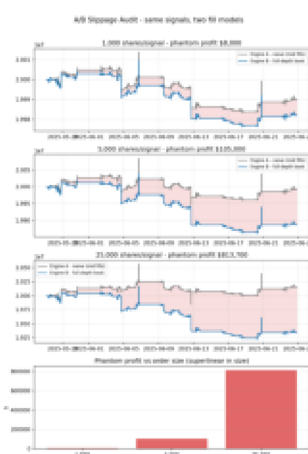
VPIN/OFI toxicity filter
robust negative: raises slippage 0.0117 vs 0.0100

Meta-labeling, judged
rejected: Engine B 0.69 -> 0.17; DSR 0.94 > 0.77

HRP vs MVO out-of-sample
MVO -0.35 collapse, HRP 0.65, 1/N still wins 0.90

FLAGSHIP: THE A/B SLIPPAGE AUDIT

Same 455 signals, two fill models - at 25k sh the naive engine's fundable Sharpe-1.9 becomes a heavy loser.



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302 C++ / 258 Python tests | 3 CI gates (build+test, format/lint, clang-tidy) | two compilers x standards x Arrow | ThreadSanitizer-clean

Whitepaper: docs/PROJECT_PHASES.md | Log: docs/TASK_BREAKDOWN.md | Walkthrough: notebooks/qse_walkthrough.ipynb